



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SWAGAT COCO KONERIPALLI

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

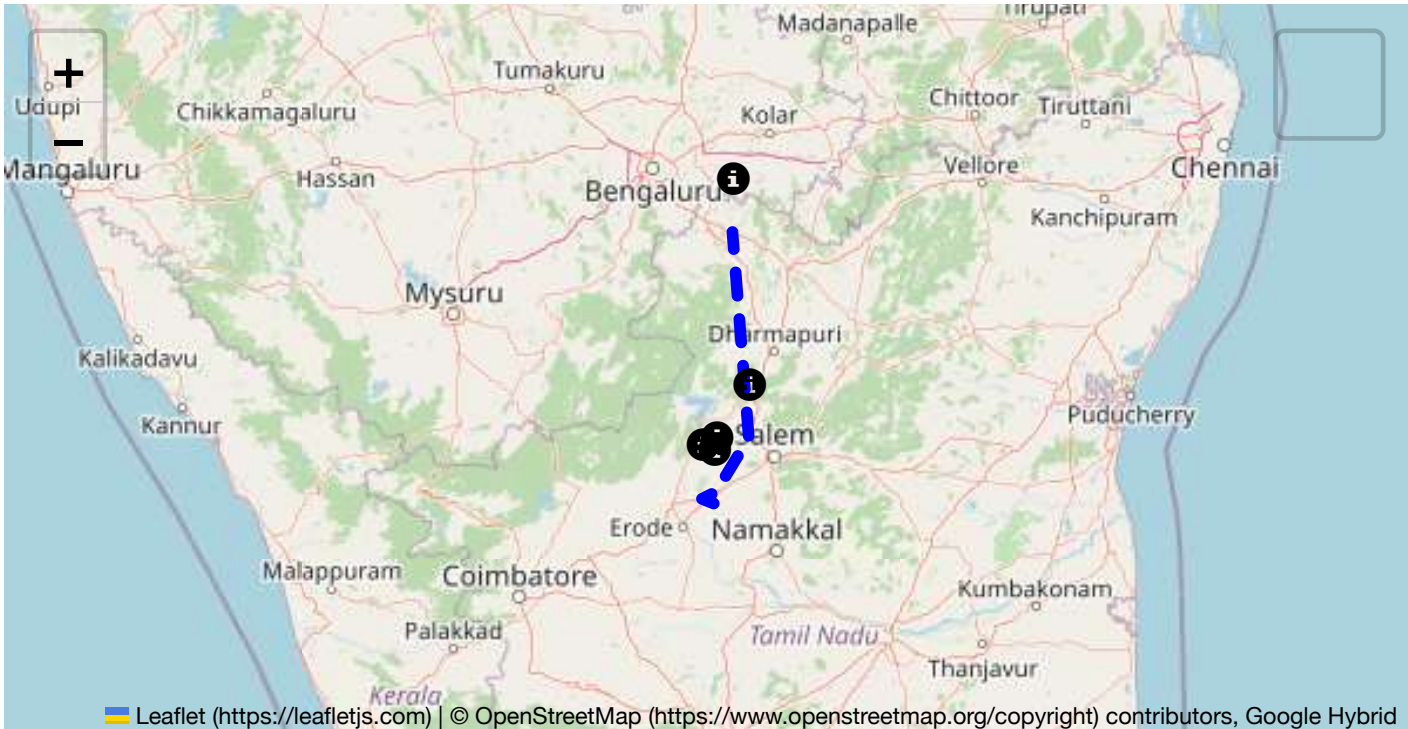
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

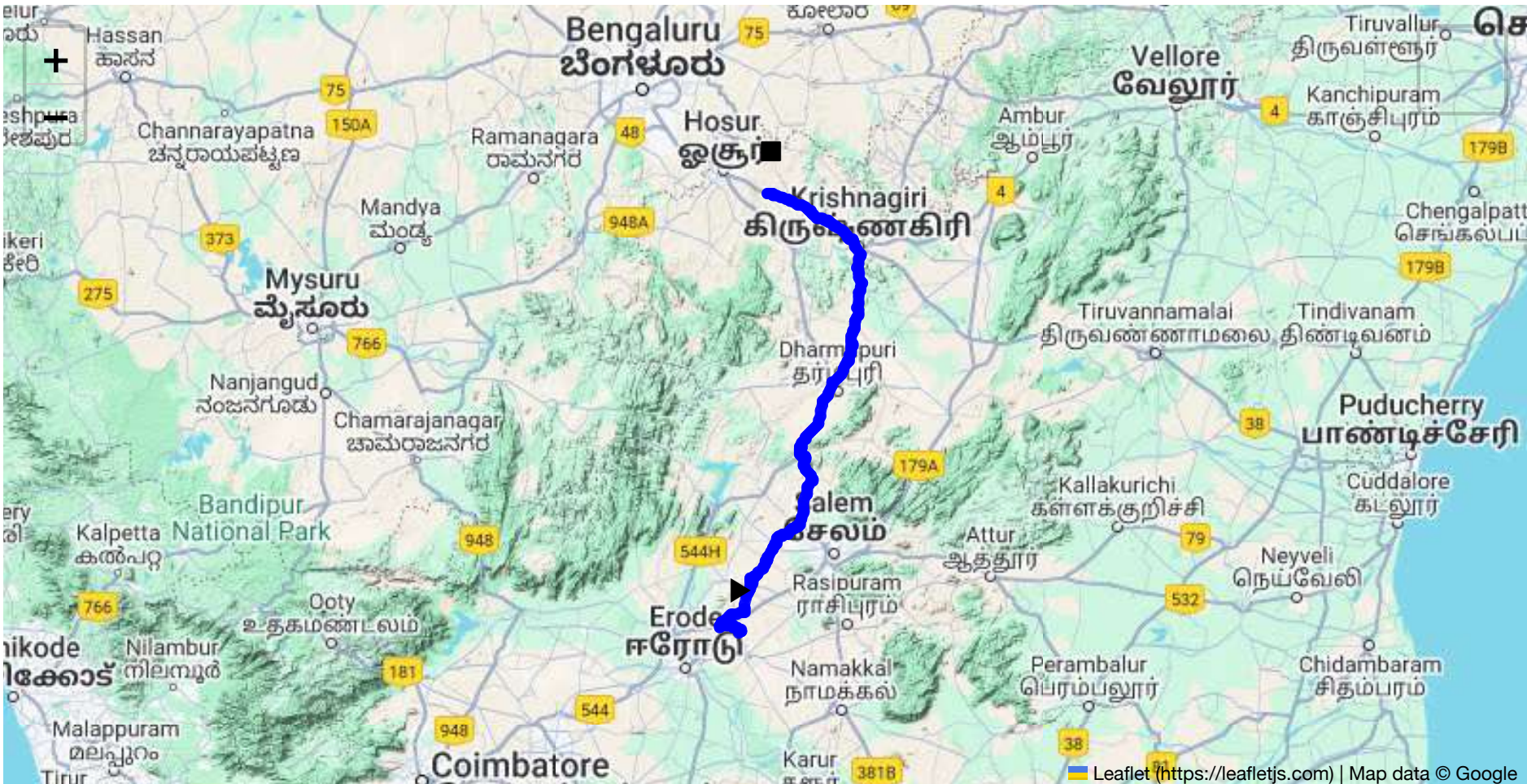
This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 186.70 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours

Start: (11.4381, 77.8734)

End: (12.679949, 77.956821)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins at Sangagiri, Tamil Nadu, and proceeds through Bhavani Main Road in Padaiveedu, SH 86 in Ayveli, and continues through Salem, en route to Addakurukki, Tamil Nadu. This path is approximately 186.70 kilometers and weaves through both rural areas and small urban centers, interspersed with state highways.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu experiences a tropical climate with high temperatures and humidity. The northeast monsoon between October and December can bring heavy rainfall, potentially causing waterlogging and slippery road conditions. Monitor weather forecasts for storms or heavy rainfall that could increase the risk of accidents.

3. Analysis of Traffic Patterns

The route encounters variable traffic volumes, with peak hours typically in the morning (8-10 AM) and evening (5-8 PM), particularly around Salem due to its urban nature. Congestion tends to build near agricultural areas during harvest seasons, as local produce is transported to markets.

4. Assessment of Road Quality and Infrastructure

While major roads like SH 86 are generally well-maintained, interior roads can be narrow with occasional potholes. Bridge maintenance and signposting improvements are ongoing, but drivers should remain vigilant for unexpected road conditions, particularly in rural stretches.

5. Suggestions for Alternative Routes for Emergencies

In case of an emergency, state highways like SH 20 and adjacent smaller roads can provide detours. It’s crucial to have GPS navigation ready to re-route efficiently and access local assistance for real-time advice.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires adherence to strict guidelines, including proper labeling, use of safety equipment, and specific travel times to avoid peak traffic. Compliance with the Motor Vehicles Act and CMVR regulations is mandatory.

7. Overview of Historical Incidents

There have been occasional incidents involving heavy vehicles in Salem due to its congestion and urban layout, typically involving minor accidents. Major hazardous material incidents are rare but historically involve inadequate handling or preventive measures.

8. Environmental Considerations and Sensitive Areas

The route passes through environmentally sensitive zones, including agricultural and forested areas. Spills or leaks could seriously affect these ecosystems, necessitating rigorous adherence to environmental protection protocols.

9. Analysis of Communication Coverage

Mobile coverage is robust in urban areas like Salem but can be patchy in rural regions and certain stretches between Salem and Addakurukki. Prepare for potential dead zones by planning stops in urbanized areas for communication needs.

10. Estimated Emergency Response Times

Response times vary: around 15-30 minutes in urban areas like Salem; up to 45 minutes to an hour in rural stretches where medical facilities and rescue services are less accessible.

12. Overall Summary of Risk Assessment

Overall, the route is moderately safe for transporting hazardous materials, provided drivers adhere to weather advisories, peak hour schedules, and regulatory standards. Preparedness for emergency detours and communication challenges is essential. Vigilance regarding road quality and environmental impact is important to mitigate potential risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	12.6797949, 77.9555036	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	Medium	11.45353, 77.85697	30 KM/Hr
5	Turn	Medium	11.45781, 77.85203	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	Medium	11.45546, 77.81800	30 KM/Hr
8	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
9	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
10	Turn	High	11.49469, 77.88488	15 KM/Hr
11	Turn	High	11.55618, 77.89789	15 KM/Hr
12	Turn	Medium	11.55699, 77.89986	30 KM/Hr
13	Turn	High	11.58228, 77.90841	15 KM/Hr
14	Turn	High	11.58340, 77.90637	15 KM/Hr
15	Turn	Medium	11.59201, 77.90649	30 KM/Hr
16	Turn	Medium	11.63882, 77.94859	30 KM/Hr
17	Turn	High	11.64226, 77.95200	15 KM/Hr
18	Turn	Medium	11.64716, 77.95118	30 KM/Hr
19	Turn	Medium	11.67868, 77.96836	30 KM/Hr
20	Turn	High	11.70223, 77.97860	15 KM/Hr
21	Turn	Medium	11.71402, 77.99586	30 KM/Hr
22	Turn	Medium	11.71704, 77.99948	30 KM/Hr
23	Turn	Medium	11.71918, 78.01127	30 KM/Hr
24	Turn	Medium	11.72708, 78.02197	30 KM/Hr
25	Turn	High	11.73769, 78.03451	15 KM/Hr
26	Turn	High	11.73651, 78.03871	15 KM/Hr
27	Turn	Medium	11.74143, 78.04613	30 KM/Hr
28	Turn	Medium	11.86847, 78.08367	30 KM/Hr
29	Turn	Medium	11.87942, 78.08197	30 KM/Hr
30	Turn	Medium	11.91402, 78.05884	30 KM/Hr
31	Turn	Medium	11.92585, 78.06507	30 KM/Hr
32	Turn	Medium	11.93122, 78.06489	30 KM/Hr
33	Turn	Medium	11.93447, 78.06305	30 KM/Hr
34	Turn	Medium	11.93704, 78.05402	30 KM/Hr
35	Turn	Medium	11.95664, 78.04781	30 KM/Hr
36	Turn	High	11.95818, 78.04978	15 KM/Hr
37	Turn	Medium	11.95988, 78.04948	30 KM/Hr
38	Turn	High	11.96399, 78.05135	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	11.96496, 78.04988	30 KM/Hr
40	Turn	High	11.96675, 78.04939	15 KM/Hr
41	Turn	Medium	12.16983, 78.16769	30 KM/Hr
42	Turn	Medium	12.19881, 78.18157	30 KM/Hr
43	Turn	Medium	12.22459, 78.19136	30 KM/Hr
44	Turn	Medium	12.24731, 78.19601	30 KM/Hr
45	Turn	Medium	12.25060, 78.19977	30 KM/Hr
46	Turn	Medium	12.25329, 78.20039	30 KM/Hr
47	Turn	Medium	12.29110, 78.19909	30 KM/Hr
48	Turn	Medium	12.39904, 78.21764	30 KM/Hr
49	Turn	Medium	12.40156, 78.21845	30 KM/Hr
50	Turn	Medium	12.40589, 78.22275	30 KM/Hr
51	Turn	Medium	12.42428, 78.22671	30 KM/Hr
52	Turn	Medium	12.44389, 78.21824	30 KM/Hr
53	Turn	Medium	12.50166, 78.22333	30 KM/Hr
54	Turn	Medium	12.50553, 78.22318	30 KM/Hr
55	Turn	Medium	12.59749, 78.13593	30 KM/Hr
56	Turn	Medium	12.63696, 78.07061	30 KM/Hr
57	Turn	Medium	12.67143, 77.97400	30 KM/Hr
58	Blind Spot	Blind Spot	12.67971, 77.95592	10 KM/Hr
59	Turn	Medium	12.67974, 77.95631	30 KM/Hr
60	Turn	High	12.68003, 77.95664	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
12	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
15	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
16	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
17	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
18	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
23	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
24	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
25	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
30	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
31	clinic	Shri Narayana Bone & Joint Clinic	12.523617, 78.210613	30 km/h	Medium
32	clinic	AZ Family Clinic	12.5234942, 78.2127422	30 km/h	Medium
34	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
35	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
36	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
27	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
28	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
33	school	Vivekananda School	12.5231937, 78.2121429	30 km/h	Medium
37	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
38	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
39	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium
40	school	New Little Flower Matriculation Higher Secondary School	12.6607923, 78.0169012	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn
Risk Level: High
Speed Limit: 10 KM/Hr
Coordinates: 12.6797949, 77.9555036



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71704, 77.99948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72708, 78.02197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74143, 78.04613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91402, 78.05884



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93122, 78.06489



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93447, 78.06305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95664, 78.04781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.42428, 78.22671



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.44389, 78.21824



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50166, 78.22333



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50553, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59749, 78.13593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.63696, 78.07061



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67143, 77.97400



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.67971, 77.95592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.67974, 77.95631



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.68003, 77.95664

